

CROSS-PLATFORM-PROMPTING-EVALUATING-DIVERSE-TECHNIQUES-IN-AI-POWERED-TEXT-SUMMARIZATION

Zero-Shot Prompting

Prompt:

Summarize the following 500-word technical article on “The Basics of Blockchain Technology” for undergraduate students in simple language. Highlight key concepts, advantages, applications, and limitations in around 250 words.

By ChatGPT

Response:

Summary of “The Basics of Blockchain Technology”

Blockchain technology is a secure digital system used to store and share information across multiple computers. It works as a distributed ledger where data is stored in blocks connected in sequence, forming a chain. Each block contains transaction data, a timestamp, and a unique hash value that links it to the previous block. This structure makes the data difficult to alter and increases security.

One of the main features of blockchain is decentralization. Unlike traditional systems controlled by a single organization, blockchain networks are managed by many computers called nodes. This improves transparency and reduces the risk of data manipulation. Blockchain also uses consensus mechanisms such as Proof of Work (PoW) and Proof of Stake (PoS) to verify and approve transactions securely.

Blockchain technology is widely known for supporting cryptocurrencies like Bitcoin, but its uses go beyond digital money. It is applied in banking, healthcare, supply chain management, voting systems, and smart contracts. Smart contracts are automated agreements that execute actions when specific conditions are met, reducing paperwork and the need for intermediaries.

The major advantages of blockchain include high security, transparency, reliability, and faster transactions. Since data is encrypted and shared across many systems, it is difficult for hackers to tamper with records.

However, blockchain also has some limitations. Some blockchain systems consume large amounts of energy, especially those using Proof of Work. Scalability and slower transaction speeds can also be challenges. In addition, legal regulations and adoption issues still exist in many countries.

Overall, blockchain technology is transforming digital systems by providing secure, transparent, and decentralized ways to manage information and transactions.

By Gemini AI

Response:

Understanding Blockchain Technology

Think of a **blockchain** as a digital ledger shared across a vast network of computers. Unlike a private bank statement, this ledger is public, synchronized, and nearly impossible to tamper with.

Key Concepts

- **Blocks and Chains:** Data is bundled into "blocks." Once a block is full, it is "chained" to the previous one using a unique digital fingerprint called a **hash**.
- **Decentralization:** There is no central authority. Every computer in the network (node) holds an identical copy of the entire history.
- **Consensus:** Before adding a block, the network must agree it is valid through specific rules, preventing fraud.
- **Immutability:** Once data is recorded, changing it would require altering every subsequent block across the entire network—a feat that is mathematically unfeasible.

Advantages & Limitations

Blockchain offers **unmatched transparency** and **security**, as encryption protects data from hackers. By removing middlemen (like clearing houses), it can also reduce costs. However, it faces **scalability** issues, where large networks struggle to process transactions quickly. Additionally, certain types of blockchain security require **high energy consumption**.

Real-World Applications

Beyond **cryptocurrency**, blockchain powers **Smart Contracts**—self-executing programs that trigger when conditions are met. It is also used in **Supply Chain Management** to track products from factory to front door, and in **Secure Voting** to ensure every vote is permanent and verifiable.

By Copilot

Response:

Blockchain Technology: A Simple Overview

Blockchain is a digital system for recording information in a way that makes it very hard to change or hack. Instead of storing data in one place, it keeps records across many computers (called nodes). Each record, or “block,” is linked to the previous one, forming a secure “chain.” This structure ensures transparency and trust without needing a central authority.

Key Concepts:

- **Decentralization:** No single company or government controls the data; it's shared across many participants.
- **Immutability:** Once data is added, it cannot be easily altered.
- **Transparency:** Everyone in the network can see the transactions.
- **Security:** Cryptography protects the data and verifies transactions.

✓ Advantages:

- Reduces fraud and tampering.
- Cuts out middlemen, lowering costs.
- Builds trust among strangers through shared records.
- Enables faster, more efficient transactions.

🔗 Applications:

- **Finance:** Cryptocurrencies like Bitcoin and secure payment systems.
- **Supply Chains:** Tracking goods from production to delivery.
- **Healthcare:** Safely storing patient records.
- **Voting:** Creating tamper-proof digital election systems.
- **Smart Contracts:** Self-executing agreements coded into the blockchain.

⚠ Limitations:

- High energy use, especially in systems like Bitcoin.
- Scalability issues—handling large numbers of transactions is still slow.
- Regulatory uncertainty in many countries.
- Complex technology that's hard for beginners to understand.

By Perplexity AI

Response:

Here's a clear 250-word summary of "The Basics of Blockchain Technology" written for undergraduates, with key concepts, advantages, applications, and limitations.

Blockchain is a distributed digital ledger that records transactions in linked blocks so data is permanent and tamper-resistant.

Each block contains transaction data, a timestamp, and a cryptographic hash that links it to the previous block; this chain structure plus cryptography makes altering past records extremely difficult. Nodes (computers) in the network hold copies of the ledger and reach agreement using a consensus mechanism (e.g., proof-of-work or proof-of-stake), removing the need for a central authority. Smart contracts are programmable rules stored on the chain that automatically execute when conditions are met.

Advantages:

- Decentralization reduces single points of failure and reliance on intermediaries, increasing trust among participants.
- Immutability and cryptographic security protect data integrity and make tampering detectable.
- Transparency and traceability let participants audit histories and follow assets across transactions.
- Automation via smart contracts cuts administrative overhead and speeds up processes.

Applications:

- Cryptocurrencies and token systems for value transfer.
- Supply-chain tracking to verify provenance and reduce fraud.
- Identity management and secure recordkeeping (e.g., land registries, academic credentials).
- Financial services (cross-border payments, settlement), voting systems, and automated legal or business workflows.

Limitations:

- Scalability and throughput can be low, causing slow or costly transactions on some networks.
- Energy use (especially older proof-of-work systems) and operational costs can be high.
- Immutability makes erroneous or fraudulent transactions hard to reverse.

- Legal, regulatory, and governance frameworks are still developing, and smart-contract bugs create security risks.

Example: a supply-chain blockchain can show each handoff of a product from factory to store, improving trust and simplifying recalls.

Few-Shot Prompting

Prompt:

Using the style of the examples above, summarize the article “The Basics of Blockchain Technology” in clear and concise language suitable for undergraduate students.

By ChatGPT

Response:

Blockchain technology is a digital system used to store and share information securely across many computers. It works through a chain of blocks, where each block contains data, a timestamp, and a unique hash value that connects it to the previous block. This structure makes the data secure and difficult to modify.

One important feature of blockchain is decentralization. Instead of being controlled by one organization, the information is managed by multiple computers called nodes. Blockchain also uses consensus mechanisms such as Proof of Work and Proof of Stake to verify transactions and maintain trust in the network.

Blockchain is mainly known for cryptocurrencies like Bitcoin, but it is also used in banking, healthcare, supply chain management, voting systems, and smart contracts. Smart contracts automatically execute agreements when conditions are met, reducing the need for intermediaries.

The main advantages of blockchain are security, transparency, reliability, and faster digital transactions. However, it also

has limitations such as high energy consumption, scalability issues, slower processing speed, and legal challenges.

Overall, blockchain technology provides a secure and transparent way to manage digital information and is becoming increasingly important in many industries.

By Gemini AI

Response:

Understanding Blockchain Technology

Think of a **blockchain** as a digital ledger shared across a vast network of computers. Unlike a private bank statement, this ledger is public, synchronized, and nearly impossible to tamper with.

Key Concepts

- **Blocks and Chains:** Data is bundled into "blocks." Once a block is full, it is "chained" to the previous one using a unique digital fingerprint called a **hash**.
- **Decentralization:** There is no central authority. Every computer in the network (node) holds an identical copy of the entire history.
- **Consensus:** Before adding a block, the network must agree it is valid through specific rules, preventing fraud.
- **Immutability:** Once data is recorded, changing it would require altering every subsequent block across the entire network—a feat that is mathematically unfeasible.

Advantages & Limitations

Blockchain offers **unmatched transparency** and **security**, as encryption protects data from hackers. By removing middlemen (like clearing houses), it can also reduce costs. However, it faces **scalability** issues, where large networks struggle to process transactions quickly. Additionally, certain types of blockchain security require **high energy consumption**.

Real-World Applications

Beyond **cryptocurrency**, blockchain powers **Smart Contracts**—self-executing programs that trigger when conditions are met. It is also used in **Supply Chain Management** to track products from factory to front door, and in **Secure Voting** to ensure every vote is permanent and verifiable.

By Copilot

Response:

Blockchain Technology: A Simple Guide

Blockchain is a digital way of recording information that makes data secure, transparent, and very hard to change. Instead of one central database, it spreads records across many computers. Each “block” of data is linked to the previous one, forming a “chain” that everyone in the network can trust.

Key Concepts:

- **Decentralization:** No single authority controls the system; it's shared by many users.
- **Immutability:** Once added, data cannot be easily altered.
- **Transparency:** Transactions are visible to all participants.
- **Security:** Cryptography ensures safe and verified records.

Advantages:

- Prevents fraud and tampering.
- Cuts out middlemen, reducing costs.
- Builds trust among strangers.
- Enables faster, more efficient transactions.

Applications:

- **Finance:** Cryptocurrencies like Bitcoin and secure payments.
- **Supply Chains:** Tracking goods from factory to customer.
- **Healthcare:** Protecting patient records.
- **Voting:** Creating tamper-proof election systems.
- **Smart Contracts:** Self-executing agreements coded into the blockchain.

Limitations:

- High energy use in some systems.
 - Scalability issues—handling many transactions is still slow.
 - Unclear regulations in many countries.
 - Complex technology that's hard for beginners to grasp.
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By perplexity

Response:

Blockchain is a distributed digital ledger that records transactions in linked blocks so data becomes permanent and hard to change. Each block holds transaction details, a timestamp, and a cryptographic hash that links it to the previous block, creating a tamper-resistant chain. Network participants (nodes) keep copies of the ledger and use a consensus method (for example, proof-of-work or proof-of-stake) to agree on which new blocks are valid, removing the need for a central authority. Smart contracts are self-executing programs stored on the chain that run automatically when pre-set conditions are met.

Advantages:

- Decentralization reduces single points of failure and lowers dependence on intermediaries, increasing trust among users.
- Immutability and cryptographic integrity make data tampering difficult and detectable.

- Transparency and traceability let anyone with access audit transaction history and follow asset flows.
- Automation with smart contracts speeds processes and reduces paperwork and manual reconciliation.

Applications:

- Cryptocurrencies and digital tokens for transferring value.
- Supply-chain tracking to verify provenance and quickly identify recalls.
- Secure recordkeeping for identities, land titles, diplomas, and medical records.
- Financial services (cross-border payments, settlement), transparent voting systems, and automated business workflows.

Limitations:

- Many blockchains face scalability and throughput limits, causing slow or costly transactions on busy networks.
- Some consensus methods (notably older proof-of-work) use large amounts of energy and can be expensive to run.
- The ledger's permanence makes reversing mistakes or fraud difficult.
- Legal, regulatory, and governance frameworks are still evolving, and smart-contract bugs can cause security vulnerabilities.

Example: In a supply chain, a blockchain can record each transfer from factory to retailer so buyers and regulators can verify origin and reduce fraud.

Chain-of-Thought Prompting

Prompt:

Read the article on “The Basics of Blockchain Technology” and summarize it step by step:

1. Identify the definition of blockchain technology.
2. Explain how blocks and hashes work.
3. Describe decentralization and consensus mechanisms.
4. Mention important applications such as cryptocurrency and smart contracts.
5. Discuss advantages and limitations.
6. Combine these points into a clear summary for undergraduate students.

By ChatGPT**Response:****Step-by-Step Summary of “The Basics of Blockchain Technology”****1. Definition of Blockchain Technology**

Blockchain technology is a digital system used to securely store and share information across multiple computers. It works like a distributed ledger where records are maintained in a transparent and secure manner.

2. How Blocks and Hashes Work

Information in blockchain is stored in units called blocks. Each block contains transaction data, a timestamp, and a unique code called a hash. The hash connects one block to another, forming a chain. If any data is changed, the hash also changes, making tampering easy to detect.

3. Decentralization and Consensus Mechanisms

Blockchain is decentralized, meaning no single organization controls the system. Instead, many computers called nodes manage the network

together. To verify transactions, blockchain uses consensus mechanisms such as Proof of Work (PoW) and Proof of Stake (PoS), which help maintain trust and security.

4. Applications of Blockchain

Blockchain is widely used in cryptocurrencies like Bitcoin. It is also applied in banking, healthcare, supply chain management, voting systems, and smart contracts. Smart contracts are self-executing agreements that automatically perform actions when conditions are met.

5. Advantages and Limitations

The major advantages of blockchain are security, transparency, reliability, and reduced dependency on intermediaries. However, it also has limitations such as high energy consumption, scalability issues, slower transaction speeds, and legal or regulatory challenges.

6. Final Combined Summary

Blockchain technology is a secure and decentralized system for storing digital information. By linking blocks with hashes and verifying transactions through consensus mechanisms, blockchain ensures transparency and data security. It supports applications like cryptocurrency, smart contracts, and supply chain management. Although it offers many benefits such as reliability and security, challenges like energy usage and scalability still exist. Overall, blockchain is becoming an important technology in modern digital systems.

By Gemini AI

Response:

Based on the core principles of blockchain technology, here is a step-by-step summary designed for an undergraduate audience.

1. The Definition

Blockchain is a distributed, digital ledger technology that records transactions across many computers. Unlike a standard database controlled by one entity, it is decentralized, meaning the data is transparent, synchronized, and immutable (cannot be changed once recorded).

2. Blocks and Hashes

Data is organized into **blocks**. Each block contains a batch of transactions and a unique digital fingerprint called a **hash**.

- **The Chain:** Each block also contains the hash of the *previous* block.
- **Security:** If a single piece of data in an old block is altered, its hash changes, breaking the link to all subsequent blocks. This makes tampering immediately obvious to the entire network.

3. Decentralization and Consensus

- **Decentralization:** There is no "master server." Instead, every participant (node) has a copy of the entire ledger.
- **Consensus Mechanisms:** To add new data, the network must agree on its validity. Mechanisms like **Proof of Work** or **Proof of Stake** ensure that all nodes reach a mathematical agreement, preventing anyone from "double-spending" or entering fake data.

4. Key Applications

- **Cryptocurrency:** The first use case (e.g., Bitcoin), allowing peer-to-peer financial transfers without a bank.
- **Smart Contracts:** Programmable "if/then" agreements. For example, a contract could automatically release payment to a vendor the moment a delivery is scanned at a warehouse.

5. Advantages and Limitations

- **Advantages:** High security, transparency, and the removal of expensive middlemen (like brokers or clearing houses).

- **Limitations:** High energy consumption for certain networks, difficulty in "scaling" to handle millions of instant transactions, and the steep learning curve for new users.
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6. The Final Summary

For undergraduate students, **Blockchain** represents a shift from "trusting a central authority" to "trusting a distributed network." By grouping data into **blocks** secured by cryptographic **hashes**, the system creates a permanent, unchangeable record.

Because it operates via **decentralization**, no single person can shut it down or cheat the system; the network relies on **consensus** to verify truth. While famous for powering **cryptocurrencies**, its real potential lies in **smart contracts** and supply chain transparency. However, for blockchain to go mainstream, it must still overcome hurdles regarding its **energy efficiency** and **processing speed**.

By Copilot

Response:

Here's a brief, step-by-step summary of *The Basics of Blockchain Technology* for undergraduate students:

1. **Definition** Blockchain is a digital ledger that records transactions securely across many computers, making it hard to alter or hack.
2. **Blocks and Hashes** Data is stored in "blocks." Each block has a unique code called a "hash" that links it to the previous block, forming a secure chain.
3. **Decentralization & Consensus** No single authority controls the system. Instead, participants agree on transactions through consensus mechanisms like Proof of Work or Proof of Stake.
4. **Applications**
 - **Cryptocurrency:** Bitcoin and other digital currencies.

- **Smart Contracts:** Self-executing agreements coded into the blockchain.
- Other uses include supply chain tracking, healthcare records, and voting systems.

5. Advantages & Limitations

- **Advantages:** Transparency, security, reduced fraud, lower costs, and faster transactions.
 - **Limitations:** High energy use, scalability challenges, complex technology, and uncertain regulations.
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By Perplexity

Response:

I don't have direct access to the specific article file right now, but based on standard explanations of "The Basics of Blockchain Technology," here is a clear step-by-step summary written for undergraduate students that follows your requested structure.

1. Definition of blockchain technology
Blockchain is a distributed digital ledger that records transactions in an ordered series of blocks across a peer-to-peer network, making records tamper-resistant and visible to participants.
2. How blocks and hashes work
Each block contains a list of transactions, a timestamp, and a cryptographic hash (a unique fixed-length fingerprint) computed from the block's contents; it also stores the previous block's hash, linking blocks into a chain. If any data in a block is changed, its hash changes and the chain link breaks, so tampering is quickly detectable.
3. Decentralization and consensus mechanisms
Instead of a single central server, many nodes (computers) each keep copies of the ledger. Nodes use a consensus mechanism—such as proof-of-work (competing to solve a puzzle) or

proof-of-stake (based on stake held)—to agree which new blocks are valid before adding them to the chain. Consensus prevents double-spending and ensures the network accepts a single agreed history.

4. Important applications (cryptocurrency and smart contracts)

Cryptocurrencies use blockchain to transfer value without central banks. Smart contracts are programs stored on the blockchain that automatically execute actions when predefined conditions are met, enabling automated payments, escrow, and decentralized applications.

5. Advantages and limitations

Advantages: decentralization reduces single points of failure and intermediaries; cryptographic hashes and consensus make records immutable and auditable; smart contracts enable automation and reduced paperwork. Limitations: many blockchains face scalability and throughput limits (slow or costly transactions under load); some consensus methods are energy-intensive; immutability makes error correction difficult; legal, regulatory, and governance issues are still evolving; smart-contract bugs can cause security losses.

6. Combined clear summary for undergraduates

Blockchain is a distributed, cryptographically secured ledger made of linked blocks that stores transaction records across many nodes. Its chaining of hashes and consensus rules make tampering difficult and let participants agree on a single transaction history without a central authority. Practical uses include cryptocurrencies for value transfer, smart contracts for automated agreements, and data tracking for supply chains, identity, and records. The technology increases transparency, trust, and automation but currently faces challenges in scale, energy use (for some systems), error reversal, and legal/regulatory maturity.

Role-Based Prompting

Prompt:

You are a professor teaching undergraduate computer science students. Your task is to explain and summarize a technical article on “The Basics of Blockchain Technology” in simple, student-friendly language. Include:

- Definition of blockchain
- Working of blocks and hashes
- Decentralization and security
- Applications of blockchain
- Advantages and limitations

By ChatGPT

Response:

Summary of “The Basics of Blockchain Technology”

Blockchain technology is a digital method of storing and sharing information securely. It acts like a distributed ledger, where data is recorded across many computers instead of being stored in one central system. This makes the system more transparent and reliable.

A blockchain is made up of blocks that store information such as transactions or records. Each block contains data, a timestamp, and a unique identifier called a hash. The hash connects one block to the previous block, forming a chain. If someone tries to change the data in a block, the hash also changes, making the tampering easy to identify. This helps maintain data integrity and security.

One important feature of blockchain is decentralization. Unlike traditional systems controlled by banks or organizations, blockchain networks are managed by many computers called nodes. These nodes work together to verify transactions using consensus mechanisms such as Proof of Work (PoW) and Proof of Stake (PoS). This reduces the risk of fraud and increases trust among users.

Blockchain technology is most famous for cryptocurrencies like Bitcoin, but its applications extend to many other

fields. It is used in banking, healthcare, supply chain management, voting systems, and smart contracts. Smart contracts are automated agreements that execute actions automatically when conditions are satisfied.

The major advantages of blockchain include strong security, transparency, accuracy, and reduced dependence on intermediaries. Transactions are recorded permanently and are difficult to alter.

However, blockchain also has some limitations. Certain blockchain systems consume a large amount of energy, especially those using Proof of Work. Other challenges include slower transaction processing, scalability issues, and legal or regulatory concerns.

Overall, blockchain technology is transforming the way digital information and transactions are managed by providing a secure, transparent, and decentralized system.

By Gemini AI

Response:

Good morning, everyone! Let's break down the architecture of Blockchain Technology. Think of this not as a complex mystery, but as a simple evolution of how we keep records.

Here is the "lecture notes" summary of the core concepts:

1. What is Blockchain?

At its simplest, a blockchain is a shared digital ledger. Imagine a Google Doc where everyone in this room has "view" access, but no one has a "delete" key. It is a distributed database that records transactions across a network of computers, ensuring that the data is transparent and permanent.

2. The Anatomy: Blocks and Hashes

- **Blocks:** Think of a block as a page in a ledger. It contains data (like a transaction), a timestamp, and a unique digital fingerprint.
- **Hashes:** This fingerprint is called a hash. It's a cryptographic string created from the data inside the block.
- **The Chain:** Each block also stores the hash of the *previous* block. This creates a literal chain. If you change even one character in Block 1, its hash changes, which breaks the link to Block 2, alerting the entire network to the tampering.

3. Decentralization and Security

Unlike a bank, which has a central server (a "single point of failure"), blockchain is decentralized. Every participant (node) has a copy of the entire chain. To add a new block, the network must reach a consensus—a mathematical agreement that the transaction is valid. This makes it incredibly secure; to hack it, you'd have to overwhelm more than half of the network simultaneously.

4. Real-World Applications

While it started with Cryptocurrency (like Bitcoin), the applications go much further:

- **Smart Contracts:** Digital agreements that execute automatically when conditions are met (no lawyers needed!).
- **Supply Chain:** Tracking a product's journey from the raw material source to the store shelf to ensure authenticity.
- **Healthcare:** Securely sharing patient records between hospitals without risking data leaks.

5. Pros and Cons

The Advantages:

- **Trust:** You don't need to know the person you're dealing with; you trust the code.
- **Efficiency:** No middlemen means faster, cheaper transactions.
- **Immutability:** Once it's written, it's written in digital stone.

The Limitations:

- **Scalability:** Because every node must verify everything, it can be slower than a centralized database.
 - **Energy:** Some security methods (like Mining) require massive amounts of electricity.
 - **Human Error:** If you lose your private access key, your data or assets are gone forever—there is no "Forgot Password" support desk.
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By Copilot

Response:

Definition

Blockchain is a **digital ledger**—a special kind of database—that records transactions securely across many computers. Once information is added, it's very difficult to change, which makes it trustworthy.

Blocks and Hashes

Data is stored in units called **blocks**. Each block contains transaction details and a unique code called a **hash**. The hash links the block to the one before it, forming a chain. If someone tries to tamper with a block, the hash changes, and the chain breaks—making fraud easy to spot.

Decentralization & Security

Unlike traditional systems controlled by banks or governments, blockchain is **decentralized**. Many computers (nodes) share and verify the data. To agree on what's valid, they use **consensus mechanisms** like Proof of Work or Proof of Stake. This ensures security without needing a central authority.

Applications

- **Cryptocurrency:** Bitcoin and other digital money.
- **Smart Contracts:** Self-executing agreements coded into the blockchain.

- **Supply Chains:** Tracking goods from factory to customer.
- **Healthcare:** Protecting patient records.
- **Voting:** Creating tamper-proof election systems.

✓ **Advantages**

- Transparency and trust
- Strong security
- Reduced fraud and middlemen
- Faster, more efficient transactions

⚠ **Limitations**

- High energy use (especially Bitcoin)
- Scalability issues—handling many transactions is slow
- Complex technology for beginners
- Unclear regulations in many countries

✳ **Summary for Students**

Blockchain is a decentralized digital ledger where data is stored in linked blocks secured by hashes. It builds trust through consensus rather than central control. With applications in cryptocurrency, smart contracts, healthcare, and more, blockchain offers transparency and security. However, challenges like energy consumption, scalability, and regulation remain.

By Perplexity

Response:

Definition of blockchain

Blockchain is a distributed digital ledger that records transactions in ordered groups called blocks, stored across many computers (nodes) so the record is visible to participants and hard to

change. This design removes the need for a single central authority while preserving a single agreed history of transactions.

Working of blocks and hashes

A block bundles a set of transactions, a timestamp, and a cryptographic hash computed from that block's contents. Each block also stores the previous block's hash, linking blocks into a chain; because hashes are like digital fingerprints, any change to a block alters its hash and breaks the link, making tampering obvious.

Decentralization and security

Instead of one server controlling the ledger, many nodes each hold a copy and validate new blocks using a consensus method (for example, proof-of-work where nodes solve a puzzle, or proof-of-stake where validators are chosen by stake). Consensus ensures nodes agree on the single valid history and prevents double-spending. Security comes from cryptographic hashes, digital signatures (public/private keys), redundancy (many copies), and the economic or computational cost required to change history.

Applications of blockchain

- Cryptocurrencies: peer-to-peer digital money and token systems.
- Smart contracts: on-chain programs that automatically execute agreed rules (payments, escrow, automated workflows).
- Supply-chain tracking: provenance, anti-counterfeiting, and easier recalls.
- Identity and recordkeeping: verifiable credentials, land registries, medical records.
- Finance and clearing: cross-border payments, settlement, and transparent audit trails.

Advantages

- Removes trusted intermediaries and reduces single points of failure.
- Immutable, auditable records improve trust and traceability.
- Smart contracts automate processes and reduce paperwork.

- Distributed copies increase availability and resilience.

Limitations

- Scalability: many blockchains have limited transactions per second and can be slow or costly when busy.
- Energy and cost: some consensus methods use substantial computation and energy.
- Immutability downside: mistakes or fraudulent transactions are hard to reverse.
- Security risks: buggy smart contracts or poor key management cause losses.
- Legal and governance: regulation and standards are still evolving, creating uncertainty.

Student-friendly summary (one paragraph)

Think of blockchain as a public notebook shared across many classmates: each page (block) lists transactions and carries a digital fingerprint (hash) that ties it to the previous page, so erasing or altering ink is immediately noticed. A group decision process (consensus) decides which new page is legitimate, so no single person controls the notebook. This approach enables cryptocurrencies, automated contracts, and trustworthy tracking of goods or records, while offering transparency and fault tolerance; however, real systems must handle limits on speed, energy use, error correction, and evolving legal rules.

Evaluation of Prompting Technique + Platform for AI-Powered Text Summarization

<u>Prompt technique</u>	<u>Platform</u>	<u>Accuracy</u>	<u>Coherence</u>	<u>Simplicity</u>	<u>Speed</u>	<u>UX</u>
Zero-shot Prompting	Gemini	4	4	3	5	4
Few-shot Prompting	Copilot	5	4	4	4	4
Chain-of-thought Prompting	Gemini	4	3	3	5	5
Role-based Prompting	Perplexity	4	4	3	3	4

Overall conclusion:

The comparative evaluation of prompting techniques across different AI platforms shows that each combination has its own advantages in text summarization. However, the performance varied based on factors such as accuracy, coherence, simplicity, speed, and user experience.

Among all the combinations tested, **Few-shot Prompting with Microsoft Copilot** emerged as the **best overall combination** because it provided the highest balance of **accuracy, coherence, simplicity, speed, and user**

experience. The summaries generated were clear, well-structured, easy to understand, and highly reliable.

- **Zero-shot Prompting with Gemini** was strong in speed and quick response generation.
- **Chain-of-thought Prompting with Gemini** offered good reasoning ability and excellent user experience.
- **Role-based Prompting with Perplexity** produced context-aware outputs but was comparatively slower and less simple.

Overall, the study concludes that **Few-shot Prompting + Microsoft Copilot** is the most effective combination for AI-powered text summarization, as it delivers the best balance between quality, readability, and efficiency.
